

REAL-TIME SCORE FOLLOWING IN SHEET MUSIC IMAGES: CLOSING THE SYNTHETIC-TO-REAL GAP

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ABSTRACT

Score following tracks a live performance against its score in real time, committing to a position at every instant with no lookahead. Doing it directly on a page of printed music removes the need for any symbolic transcription, but such systems work on synthesized audio and fail on real recordings, and closing that gap is the field’s central open problem: the strongest published system tracks 95.5% of onsets within 0.5 s on synthetic audio and 88.3% on real piano recordings, and earlier systems fall much further. Every remedy proposed so far has treated the gap as a perception failure. We show it is not. On real recordings the perception model already proposes a correct position at every scored onset, and an oracle under the same real-time constraints, committing at each instant and never moving backwards, reaches 99.6% where the deployed system reaches 80.0%. We present CANDOR, which follows the score by deciding sequentially among the positions the page affords, combining a transition model over displacements with a set-based scorer that judges each position against its competitors. CANDOR tracks 94.4% of onsets within 0.5 s of real piano recordings against 88.3% for the previous best, and holds that margin across three independently trained perception models. Analysis of the remaining error shows that the image representation is irrelevant while the temporal history is decisive, that accuracy on synthesized audio barely predicts accuracy on real audio, and that the forward-motion assumption fails at repeats.

Index Terms— Score following, sheet music, audio-to-score alignment, real-time, music information retrieval

1. INTRODUCTION

A musician plays from a printed page; a machine listens, looks at the same page, and at every moment reports where in the music the performer has got to. That is score following, and it is what automatic page turning [1], live computer accompaniment [2] and score-synchronised visualisation are built on.

What makes it hard is that it happens *during* the performance: the system sees only what has been played, commits to a position immediately, and never revises it. That is the difference from offline audio-to-score alignment, where the whole

recording is in hand and a global method such as dynamic time warping settles every position at once [3]. Everything below respects this constraint, including the oracles we use to measure what is achievable.

The other requirement is the score itself. Classical followers need it symbolic and track the performance over a discrete set of score states [4, 5], but most music exists only as printed paper and optical music recognition is still too error-prone to supply that form automatically [6]. Sheet-image score following drops the requirement: the system reads a page of pixels and a stream of audio and reports a position on the page [7, 8, 9, 10, 11, 12].

The gap. Training such a system needs performances aligned to notehead coordinates on the page, an annotation that is expensive enough that essentially all of it is produced by synthesis: a score is rendered to MIDI, the MIDI to audio, and the alignment comes for free. Systems trained this way work on synthesized audio and degrade sharply on a real instrument in a real room. The published figures are stark. Between synthetic audio and real piano recordings of the same pieces, CUNet [9] falls from 85.5% of onsets tracked within 0.5 s to 22.4%, CYOLO-SB [11] from 89.3% to 79.9%, and the strongest published system CODA [12] from 95.5% to 88.3%. Every remedy proposed has been a perception remedy: reverberant augmentation [10], extra proprietary recordings [11], or a new end-to-end architecture [12].

Perception is not the bottleneck. We tested that premise before accepting it. A conditional detector for this task does not produce a position directly; it scores thousands of candidate noteheads on the page and reports the most confident one. Keeping the 256 most confident, we find that on real room-microphone recordings a candidate within the 0.5 s tolerance is present at *every one* of 4,149 scored onsets. An oracle free to pick any of them is therefore perfect, but that oracle could not be run live; the one that matters must commit at each instant and may never move backwards, exactly as a deployed follower must, and it still reaches 99.6%. The deployed rule reaches 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio survive the real-time constraint and are recoverable from what the model already saw. The room degrades the confidences, not the evidence.

This paper. We therefore treat sheet-image score following as a sequential decision over the positions the page affords,

rather than as the regression of a coordinate. CANDOR makes that decision with two parts: a transition model scoring a displacement against how much time has passed, and a set-based scorer judging each candidate against its competitors in the same instant, trained listwise on synthesized audio alone. The decision is causal and greedy, with no lookahead and no beam, and costs 1.25 ms per instant on a single CPU thread against the 50 ms available at 20 frames per second. On real piano recordings CANDOR tracks 94.4% of onsets within 0.5 s, against 88.3% for the best published system.

Our contributions are: (i) evidence, from candidate-set oracles, that this gap is a decision failure and not a perception failure; (ii) CANDOR, which acts on it and sets a new best on real recordings by a margin holding across three independently trained perception models; (iii) an analysis of what remains, covering the irrelevance of the image representation, the unreliability of synthesized audio for model selection, and the failure at discontinuities.

2. RELATED WORK

Sheet-image score following. Three approaches have been tried. *Dense localizers* treat the page as a canvas: MM-Loc [7] and CUNet [9] emit an audio-conditioned heatmap and read a coordinate off its centre of mass. *Agents* treat it as control: a reinforcement-learning policy [8] moves a reading head, rewarded for staying aligned. *Conditional detectors* treat it as a query: CYOLO [10] encodes the recent audio into a vector and runs object detection, so the audio asks which part of the page it sounds like, and CYOLO-SB [11] adds heads for the enclosing system and bar. The concurrent CODA [12] cascades system, bar and note selection over the layout regions of an annotated score, which it requires at test time.

A common perception module. Beneath these differences the field has settled on one visual front end: a convolutional network reads the page and its feature maps are modulated by feature-wise linear modulation [13], $f_{\text{FiLM}}(x) = s(z) \odot x + t(z)$, from a vector z encoding the recent audio, in CYOLO’s downscale and upscale blocks and at the deeper stages of CODA’s backbone. Our diagnosis concerns that module, and our experiments use three separately trained instances of it, so the conclusions are about the family and not one checkpoint.

3. CANDOR

3.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$. At each instant the perception model reads the audio so far and the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent.

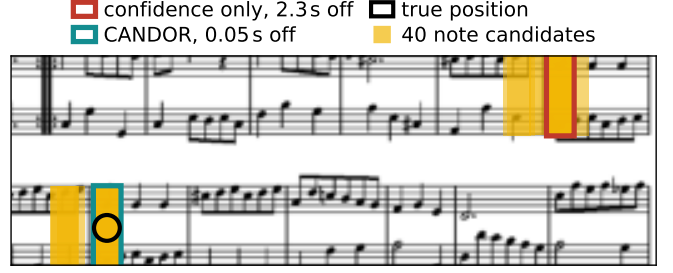


Fig. 1. One instant of a real room-microphone recording. Yellow: 40 of the candidate positions the detector scores. The most confident (red) lies on the wrong staff, 2.3 s from the truth; CANDOR (teal) takes a less confident candidate on the true note (ring). The correct position was available and was not chosen, which is the characteristic failure on real audio.

The released detectors score 8,112 boxes per instant and report

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} o_i. \quad (1)$$

CANDOR replaces Eq. 1 and reads only $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1}$.

3.2. Transition model

Successive decisions must describe a plausible trajectory. Between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (2)$$

$\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ with $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost. Combined with confidence, $h_i = \log o_i + \log \pi_i$ already recovers a large part of the gap on its own.

3.3. Set-based scorer

The transition model treats candidates independently given the history. What it cannot express is comparative: whether *this* box, among those on offer in this instant, looks like a real notehead rather than a spurious response. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$, holding its confidence relative to the instant’s best, box geometry, motion relative to the history, relation to any coarse regions the model emits, and a neighbourhood group (how many candidates fall within 25 and 100 px, spacing to the nearest on either side, rank in reading order, size against the instant’s median). We also read f_i , the 128-dimensional FiLM-conditioned feature from which the detection head scored the box, so the decision sees the representation perception used. A permutation-invariant network [14], as in learned re-scoring of detections [15], embeds

the set and pools a context shared by all candidates:

$$e_i = g([\phi_i; \text{ReLU}(Wf_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (3)$$

$$s_i = r([e_i; c]), \quad (4)$$

with g, r two-layer perceptrons (64, 32 and 64, 1 units) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is taken per dimension, so every candidate is judged against its competitors rather than in isolation.

3.4. Training and inference

The decision is supervised listwise. With ε_i the error in frames of candidate i , the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_j \exp s_j}, \quad (5)$$

so the scorer learns to order a set rather than to regress a coordinate. History features come from previous decisions, perturbed on 30% of samples by Laplace noise of scale 30 px to expose training to the drift of sequence prediction [16]. At inference CANDOR decides

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (6)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on a single CPU thread against the 50 ms available at 20 fps. Training on the residual $s_i - h_i$ and deciding $\arg \max_i (s_i + h_i)$ is Eq. 6 at $\beta = 0.5$ and eliminates all four constants.

4. EXPERIMENTAL SETUP

Data. The decision model is trained on 353 pieces of MSMD [17], synthesized and convolved with measured room impulse responses [18], and validated on 19, using AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). The evaluation set is the real-performance benchmark of [8]: 16 pieces, 25 pages and 4,149 onsets on a Yamaha AvantGrand N2 captured with a room microphone. A held-out set of 80 further pieces is rendered with a real impulse response and noise from 12 to 0.5 dB SNR.

Metric. The ratio of onsets whose predicted position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [19].

Constants. (μ, σ, J, β) are fitted leave-one-piece-out: for each evaluation piece they are chosen on the other 15, pieces weighted equally, so none influences its own decoding. The residual variant needs no fit at all. Seeds move results by about a point, so rows are seed means with deviations.

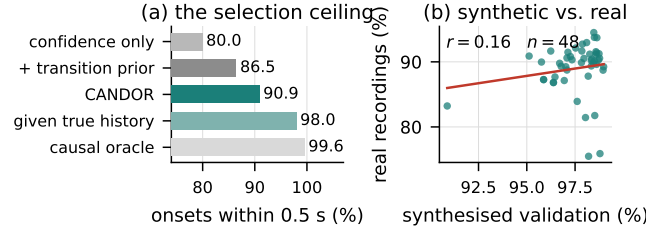


Fig. 2. (a) Every step from the deployed rule to the ceiling on the real recordings; CANDOR to *given true history* is drift, above it is ranking. (b) Each point is one of 48 trained models: accuracy on synthesised validation audio against accuracy on the real recordings.

5. RESULTS

5.1. A new best on real recordings

Table 1 sets CANDOR against the published literature on the same recordings. It tracks 94.4% of onsets within 0.5 s, where the previous best was 88.3%, and leads at every threshold. The margin is not an artifact of one perception model: retrained on whichever candidates each model emits, the decision gains 20.4, 10.7 and 7.9 points on three of them, and on the weakest, whose training data is entirely public, it passes the point at which the strongest one begins. A decision model does not need good proposals, only proposals containing the answer. On the 94 synthetic test pieces CANDOR reaches .955 where argmax reaches .893, and on 80 held-out pieces under a real impulse response and noise it beats the transition model at every level from 12 to 0.5 dB.

The tightest thresholds qualify this: on CYOLO-SB candidates CANDOR reaches .690 and .714 at 0.05 and 0.1 s, below CODA's .721 and .743, because a decision is only as precise as the boxes it chooses between. Deciding recovers which notehead, not where within it.

Table 2 separates the parts: the transition model alone, with no learned parameters, accounts for roughly half the gain, and the residual variant, which fits no constant to any evaluation data, still reaches 88.7%.

5.2. What the decision depends on

Not the image. Giving the scorer the FiLM-conditioned backbone feature, DINOv2 patch features [20] on the same grid, or no image feature at all yields 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation. On synthesized validation audio the same three differ by 3.4 points, so those features are learnable and useful there and worth nothing on real audio. **Nor the encoders.** We rebuilt the perception model with a different audio encoder twice: a frozen MERT-95M tower [21] with only a projection trained, which reaches 46.2 against 90.6 and overfits 12.6-fold on the 353 training pieces, and the two-layer causal Mamba of [12] trained from scratch as they train

Table 1. Real piano recordings: 16 pieces, 4,149 onsets, Yamaha AvantGrand N2 captured with a room microphone. Ratio of onsets tracked within each error threshold. [†] perception model trained with additional proprietary recordings, public checkpoint. *Layout*: system and bar annotations required at test time. Δ is the gain at 0.5 s over the perception model the row is built on. Published figures as reported in [12].

Method	Onset error threshold (s)					Layout	Δ
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0		
MM-Loc [7]	.364	.406	.585	.611	.735	no	
RL agent [8]	.185	.200	.476	.603	.901	no	
CUNet [9]	.113	.125	.224	.266	.443	no	
CYOLO [10]	.563	.581	.712	.749	.919	no	
CYOLO-SB [11]	.610	.630	.799	.832	.960	no	
CYOLO-SB+A [†] [11]	.682	.706	.865	.891	.981	no	
CODA [12]	.721	.743	.883	.914	.987	yes	
CANDOR on CYOLO	.726	.747	.916	.950	.991	no	+20.4
CANDOR on CYOLO-SB	.690	.714	.906	.942	.982	no	+10.7
CANDOR on CYOLO-SB+A [†]	.760	.785	.944	.969	.996	no	+7.9

Table 2. Components of CANDOR, cross-validated per piece on the real recordings at 0.5 s. *argmax* is Eq. 1 over the same candidates, *+transition* adds Eq. 2, *full* adds the scorer.

	<i>n</i>	<i>argmax</i>	<i>+transition</i>	<i>full</i>
<i>Perception model</i>				
CYOLO [10]	3	71.3	72.5	91.6 $\pm$ 1.4
CYOLO-SB [11]	6	80.0	86.5	90.6 $\pm$ 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 $\pm$ 0.9
<i>Decision model, on CYOLO-SB candidates</i>				
residual, no fitted constants	3	80.0	86.5	88.7 $\pm$ 0.9
without temporal history	3	80.0	86.5	83.7 $\pm$ 0.4
<i>Image features f_i read by the scorer</i>				
FiLM-conditioned backbone	6	80.0	86.5	90.6 $\pm$ 0.9
DINOv2 [20]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9

it, which reaches .. The two are not matched on pretraining and fail differently, one on generalisation and one on capacity, and arrive at the same place. Replacing the *visual* stem with frozen DINOv2-base patches on the identical 52×52 grid, against a control retrained under the same conditions, gives ..

The history. Removing the features computed against the previous decision costs 7.2 points, the largest single ablation here. Teacher forcing attributes two thirds of the remaining error to drift: given the true previous position the same scorer reaches 98.0% against the causal oracle’s 99.6% (Fig. 2a). The sequence is hard, not the image.

Synthesized audio does not predict real audio. Across 48 trained models, accuracy on the synthesized validation split correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b); the best-validation model scores 89.3 on real audio, below one reading no image features at all. A reverberant held-out set does better ($r = 0.80$) and still mispicks. Since the whole field selects models on synthesized audio, this is a caution beyond our own experiments, and it is why our constants are cross-validated on the target recordings instead.

Table 3. Recovery from 41 discontinuities spliced into the real recordings, on CYOLO-SB+A candidates. *rec@1*: fraction first tracked correctly within 1 s. *post*: onsets tracked within 1 s over the following 5 s.

Decision rule	break present		no break	
	rec@1	post	rec@1	post
argmax (1)	0.27	0.27	0.15	0.26
CANDOR	0.00	0.08	0.00	0.09
+ silence trigger	0.15	0.17	0.00	0.09
+ confidence trigger	0.10	0.26	0.02	0.23

5.3. Where the forward assumption fails

Eq. 2 assumes the performance moves forward, which is why the decision is tractable and also why a repeat or a jump breaks it. Following the protocol of [12] we splice three discontinuities into each performance at random destinations, re-running perception on the spliced audio so its conditioning is genuinely wrong afterwards, and excluding jumps across a page boundary where state resets anyway, leaving 41.

Table 3 reports a systematic failure. After a jump CANDOR is far worse than the rule it replaces, 0.08 against 0.27 of onsets tracked over the next five seconds, because the transition model holds the trajectory at a stale position. Suppressing it on detected silence helps where a break exists and not otherwise, as its design implies; triggering on the scorer’s own confidence needs no break and recovers most of the loss, but never beats argmax; the same ordering holds over 220 jumps on the synthetic benchmark. A transition model strong enough to be worth ten points on continuous playing is strong enough to be wrong at a discontinuity, and reconciling the two is open.

6. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a perception failure. It is a decision failure: a correct position is already among the candidates at every onset, and choosing among them in real time takes real piano recordings from 80.0% to 94.4%, against 88.3% previously.

Limitations. The correct page is given, all performances are piano, and 16 evaluation pieces cannot resolve differences of a point.

7. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

8. ACKNOWLEDGMENTS

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